COORDINATION STUDY - NOT FOR CONSTRUCTION, FABRICATION OR MATERIAL ORDERS

# Design basis and route to construction

Current Proposed model / brick cavity masonry / concrete basement concept

## What this issue establishes

An interactive structural register linked to the current architectural model. The same element geometries generate the 3D workspace, floor plans, projected elevations and section. It identifies envelope zones, modelled opening heads and named support studies.

## Owner decisions

New house walls: brick outer leaf, insulated cavity and load-bearing blockwork inner leaf. The separate garden workshop retains its 215 mm solid-masonry concept. Ground and engineering reports are not available, confirmed by the owner on 26 October 2026.

## Known architectural constraints

Current basement footprint: 115.525 m2, with its stepped courtyard edge. New wing roof: continuous, matching 29.745-degree pitches. Loft-only courtyard projection: 720 mm. The accepted original-house stair has approximately 2.01 m checked architectural headroom in places; actual structural depth and tolerances remain critical.

## Dimensions and quantity meaning

Dimensions describe model envelopes, not surveyed spans or fabrication lengths. Areas are projected architectural geometry. No whole-house brick, block, steel or concrete procurement quantities have been issued. Model fragments must first be reconciled into wall leaves, structural floor systems and designed foundations.

## What is not designed

Steel sections and connections; joist type, size, spacing and direction; masonry strengths and bearings; lintel products and padstones; lateral restraint and stability; foundations; basement reinforcement and waterproofing joints; temporary works and excavation sequence.

## Evidence required next

Commission a measured and structural survey, opening-up of existing structure and foundation investigations, geotechnical investigation with groundwater information, drainage/service records and tree constraints. The project structural engineer must set the investigation scope; trial pits and excavation need planned safe execution.

## Issue path

1. Reconcile measured geometry and decide the wall/floor assemblies.
2. Agree vertical and lateral load paths and support locations.
3. Calculate and check members, connections and foundation reactions.
4. Coordinate basement structure, waterproofing and temporary works.
5. Add the checked design to this register and produce construction drawings, reinforcement and member schedules.
6. Complete the applicable building-control process and site verification before release for construction.

## Current approval

Zero construction-approved elements. A competent structural engineer with basement experience must design/check and take responsibility for the structural scheme. These drawings do not establish safe excavation or a compliant structural solution.

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# Calculation and drawing deliverables

Checklist for the appointed structural engineer and coordinated model

## Loads and stability

Site-specific wind and snow; permanent, imposed, partition and point loads; terraces, plant and vehicle effects where applicable. Confirm current standards/UK National Annexes and design combinations. Define diaphragms, restraint, racking frames and the complete reaction path to the ground.

## Steel and opening supports

Each final member reference must link to load derivation, analysis, bending/shear/buckling checks, deflection criteria, connection design, bearings/padstones and reactions. Coordinate cavity outer-leaf support and thermal/fire/corrosion requirements. Existing openings require verification, not automatic replacement.

## Floors and roof

Choose systems after support layout: timber/I-joist/metal-web or other engineered floors as appropriate; no joist direction has been selected. Provide spans, spacing, trimming, hangers, bearing, restraint, vibration and service-zone details. Coordinate ridge/hip/dormer members, rooflights and cantilever backspan.

## Basement and foundations

Select the foundation/retaining system from geotechnical results and adjoining foundation surveys. Check earth/water/surcharge pressures, stability, settlement, flotation, bending, shear, punching and cracking as applicable. Issue concrete specifications, cover, reinforcement plans/sections, bar schedules, joints and waterstops, coordinated with the waterproofing specialist.

## Masonry and quantity take-off

Final drawings must separate brick outer leaf, cavity/insulation and block inner leaf. Include verified openings, lintels, cavity trays, closers, ties, restraint, movement joints, DPC, substructure and local bearing details. Quantify each material from coordinated assemblies, with explicit deductions, specials and waste; do not sum mesh-envelope areas.

## Construction sequence and record

Define temporary support, excavation and retaining-wall sequence, monitoring and adjoining-building protection with the designers/contractor. Record inspections and design changes. Fabrication and procurement dimensions must be checked on site. No sequence or reinforcement schedule is supplied in this P01 issue.

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# Key coordination questions

Items to resolve while protecting the accepted layout

### Q01 / House wall build-up

The visual model contains thin single wall shells. Reconcile brick/cavity/blockwork thickness before freezing room sizes or ordering lintels.

### Q02 / Basement and existing foundations

Investigate soil, groundwater, drains, tree influence and adjacent foundations. No safe excavation, retaining-wall thickness, footing size or reinforcement is established.

### Q03 / Large openings and transfer loads

Glazed courtyards, garage openings and the chamfered garden room require support and stability design. A window frame is not assumed to support the floor or masonry above.

### Q04 / Continuous roof and 720 mm projection

Coordinate ridge/hip/dormer support and cantilever backspan with a structural engineer while retaining the accepted flat-ceiling route.

### Q05 / Floor depth and stair headroom

The checked architectural headroom is only about 2.01 m in places. Engineered floor depths, finishes and tolerances may require layout adjustment.

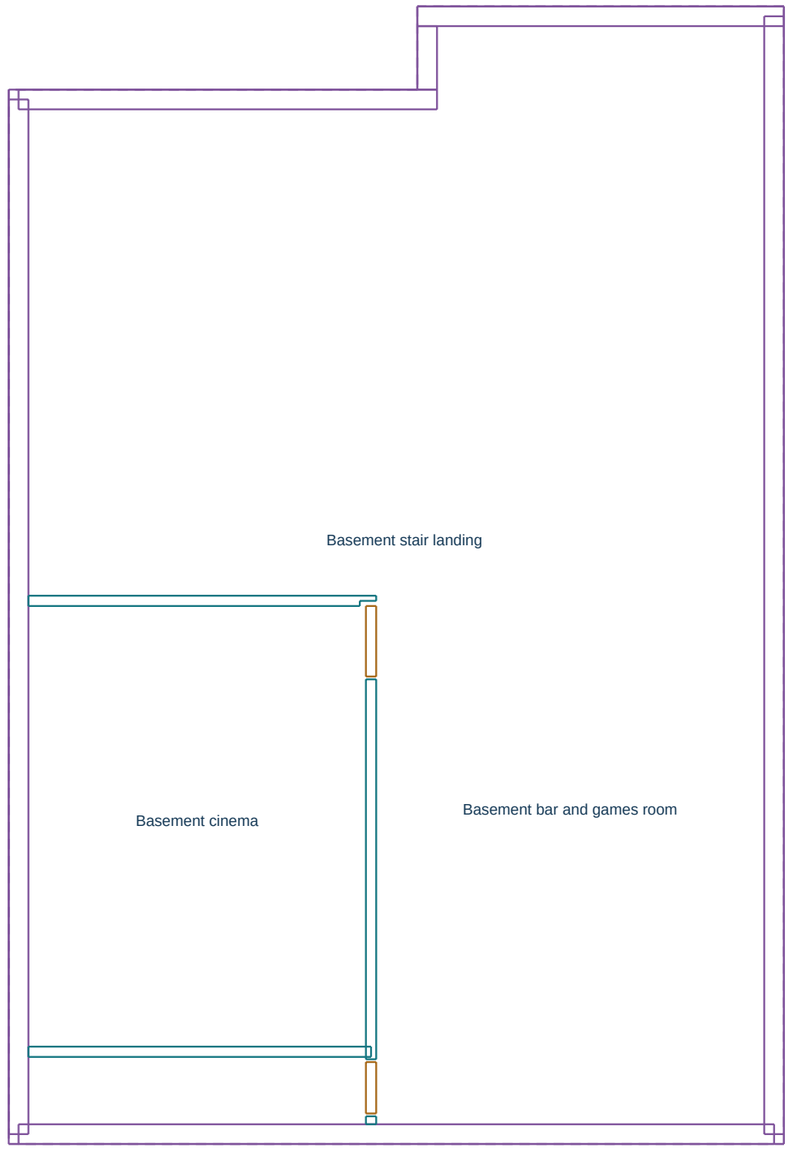
### Q06 / Quantities and completeness

Model opening heads are candidates, not a complete lintel order. Whole-house brick/block quantities await leaf geometry, verified openings, movement joints, below-ground work and waste allowances.

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# Basement coordination plan

Shared element geometry / dimensions in model metres / use the interactive register to identify lines




5 m · model geometry

## Reading this drawing

Cyan: wall envelopes.

Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

Plans cut at 1.20 m above the selected floor. Floor outlines and opening heads are projected for coordination. Hidden/interior lines remain visible.

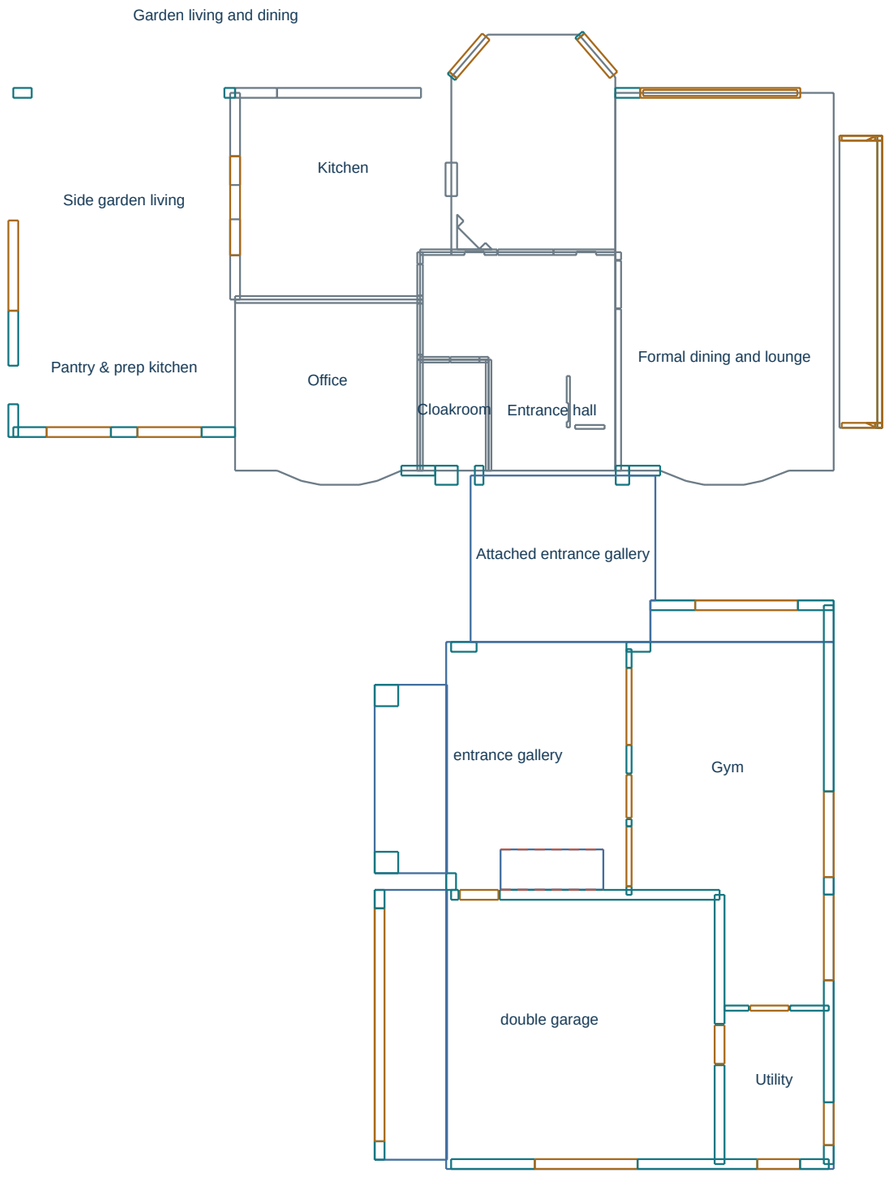
No issued steel sizes, joist spacing, lintel products, concrete thicknesses, reinforcing bars or foundation dimensions. Do not build or order from this issue.

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# Ground floor coordination plan

Shared element geometry / dimensions in model metres / use the interactive register to identify lines




5 m · model geometry

## Reading this drawing

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Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

Plans cut at 1.20 m above the selected floor. Floor outlines and opening heads are projected for coordination. Hidden/interior lines remain visible.

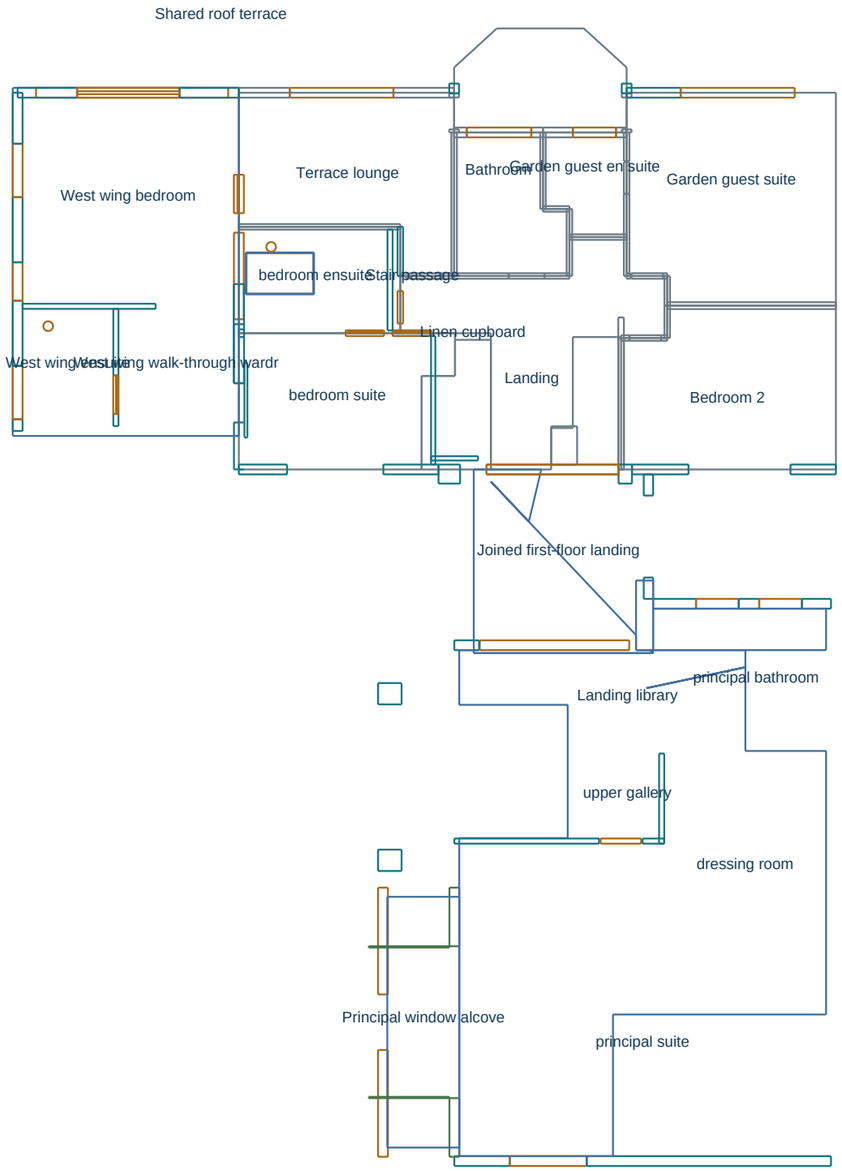
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# First floor coordination plan

Shared element geometry / dimensions in model metres / use the interactive register to identify lines




5 m · model geometry

## Reading this drawing

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Amber: modelled opening-head candidates.

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Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

Plans cut at 1.20 m above the selected floor. Floor outlines and opening heads are projected for coordination. Hidden/interior lines remain visible.

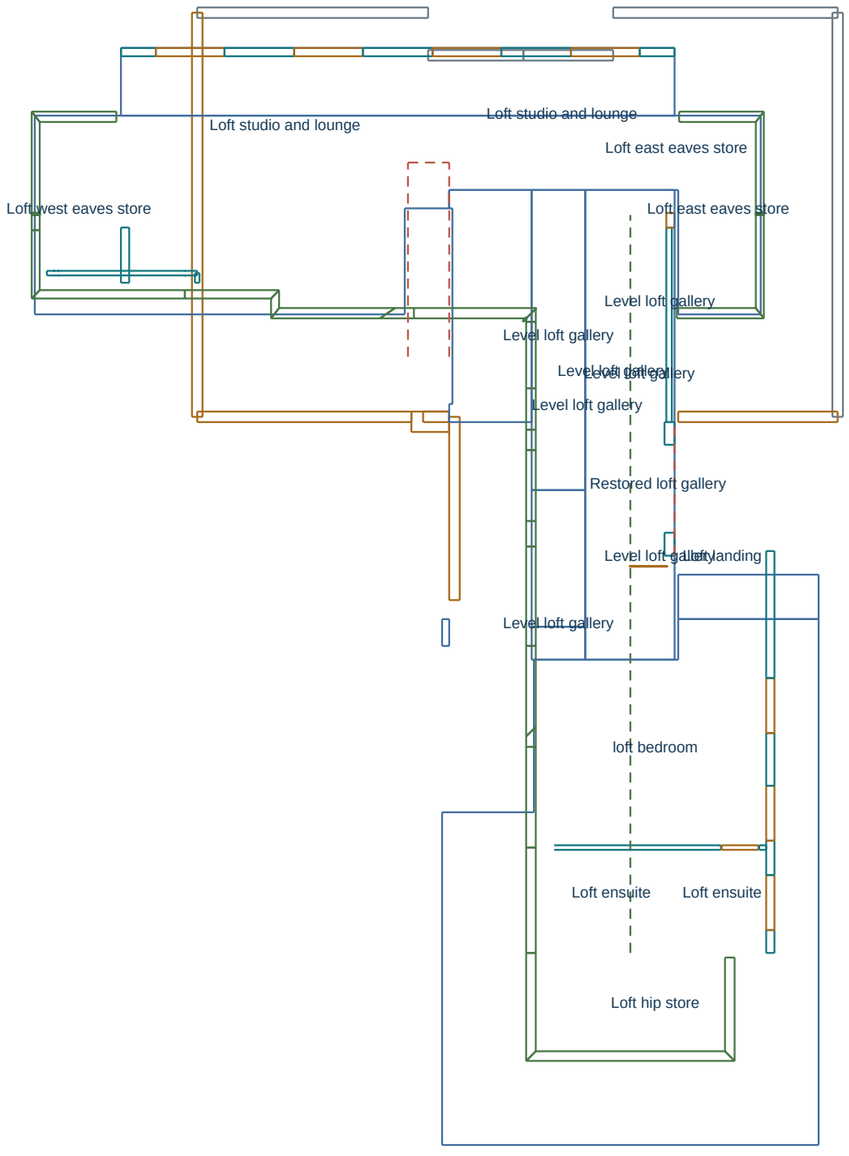
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# Loft coordination plan

Shared element geometry / dimensions in model metres / use the interactive register to identify lines




5 m · model geometry

## Reading this drawing

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Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

Plans cut at 1.20 m above the selected floor. Floor outlines and opening heads are projected for coordination. Hidden/interior lines remain visible.

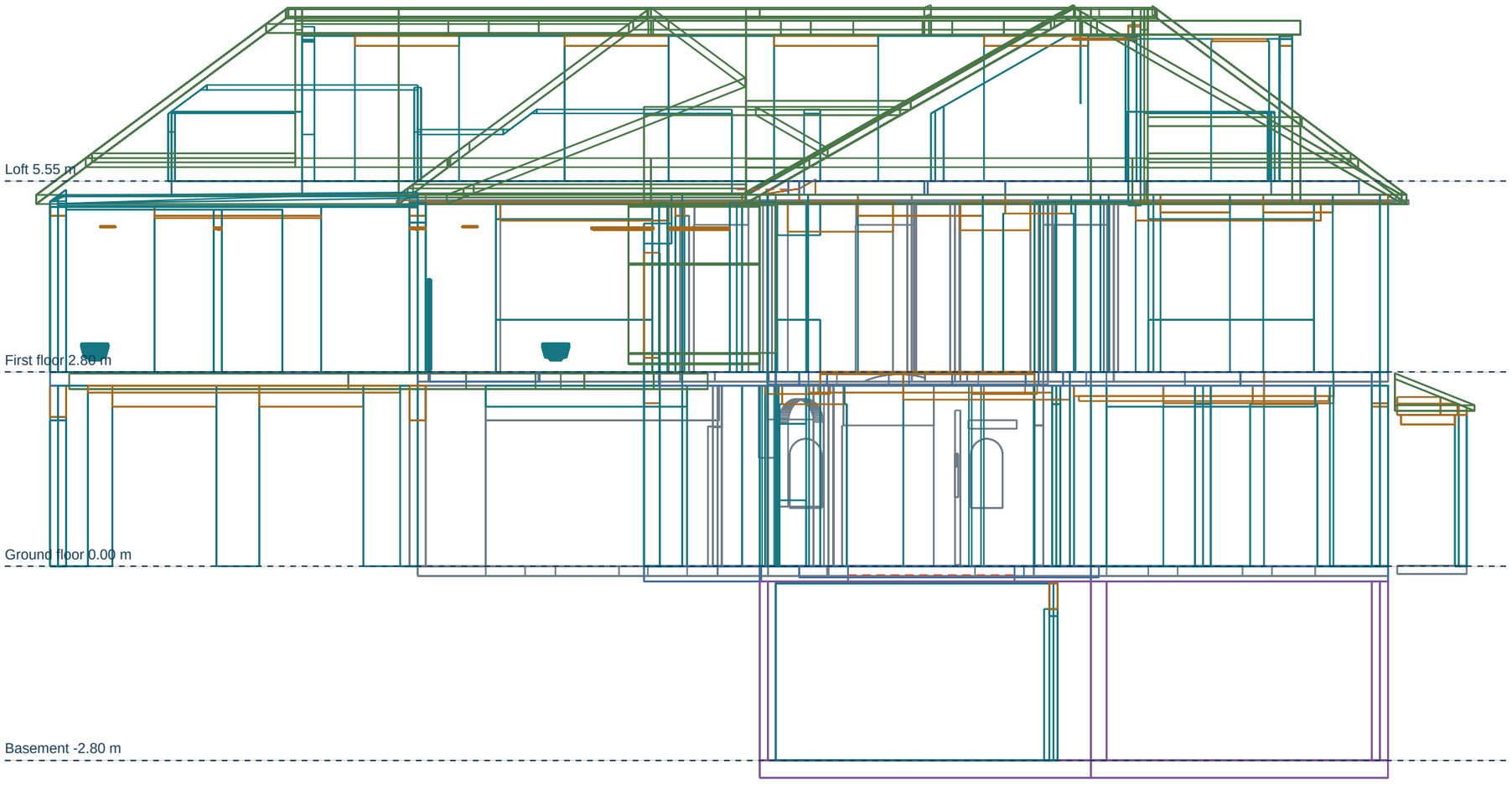
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# Front structural projection

Shared element geometry / dimensions in model metres / use the interactive register to identify lines




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Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

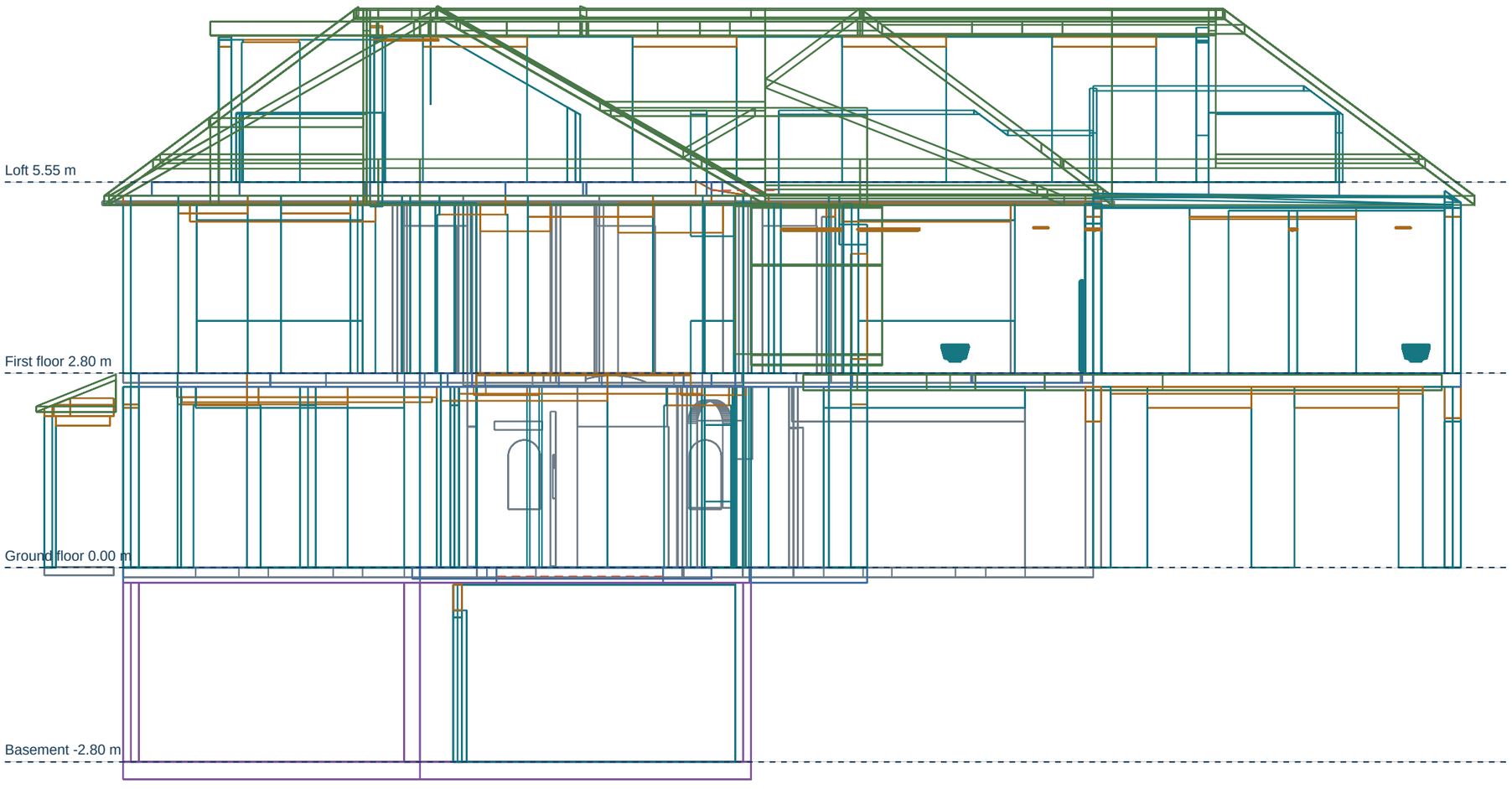
This is an orthographic structural projection with internal lines visible, not a hidden-line architectural elevation.

No issued steel sizes, joist spacing, lintel products, concrete thicknesses, reinforcing bars or foundation dimensions. Do not build or order from this issue.

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# Rear structural projection

Shared element geometry / dimensions in model metres / use the interactive register to identify lines




## Reading this drawing

Cyan: wall envelopes.

Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

This is an orthographic structural projection with internal lines visible, not a hidden-line architectural elevation.

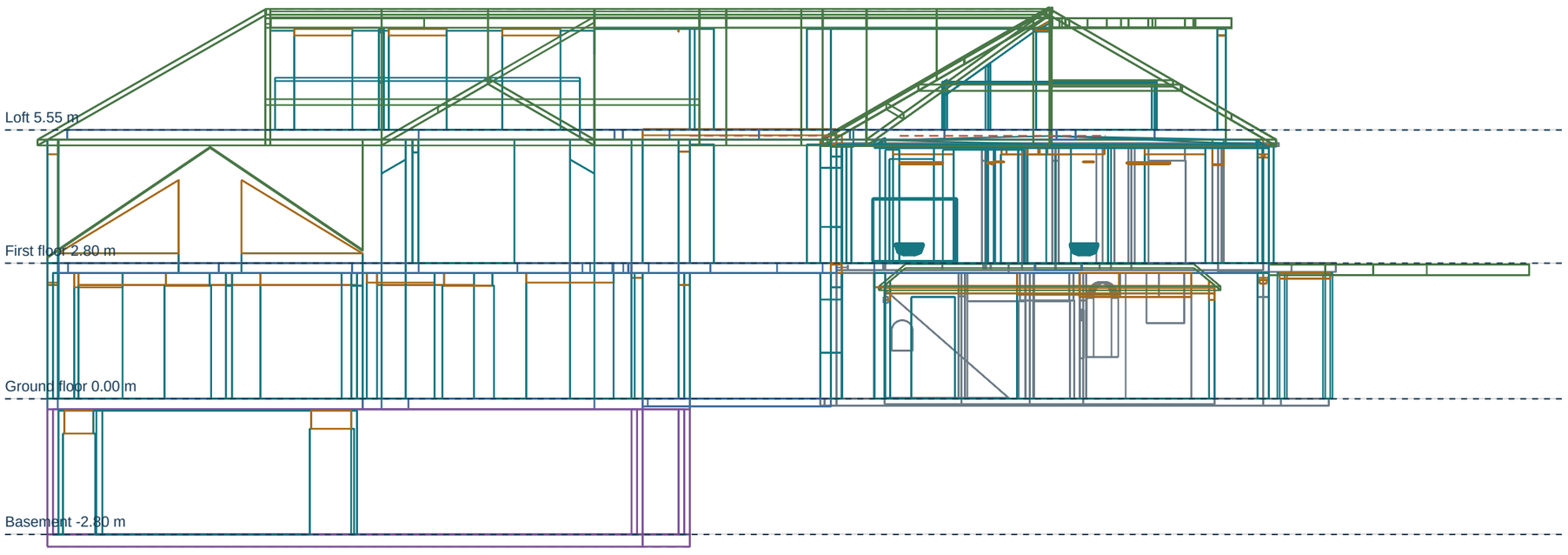
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# West structural projection

Shared element geometry / dimensions in model metres / use the interactive register to identify lines




5 m · model geometry

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## Reading this drawing

Cyan: wall envelopes.

Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

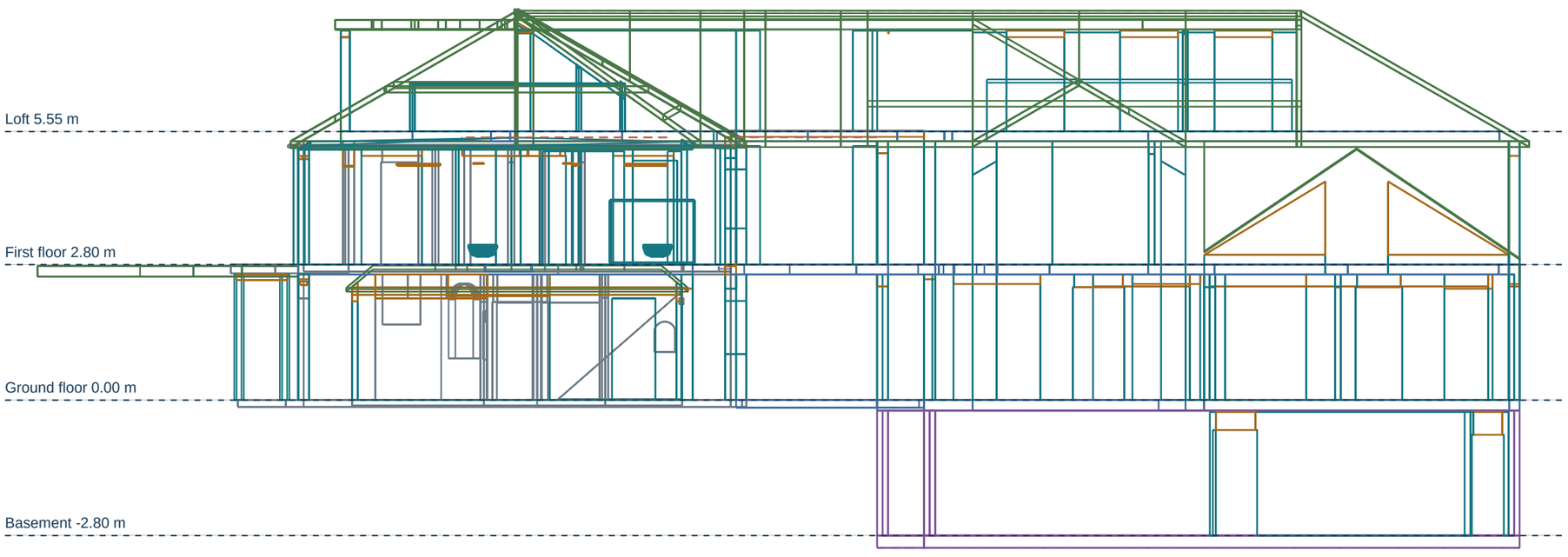
This is an orthographic structural projection with internal lines visible, not a hidden-line architectural elevation.

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# East structural projection

Shared element geometry / dimensions in model metres / use the interactive register to identify lines




5 m · model geometry

## Reading this drawing

Cyan: wall envelopes.

Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

This is an orthographic structural projection with internal lines visible, not a hidden-line architectural elevation.

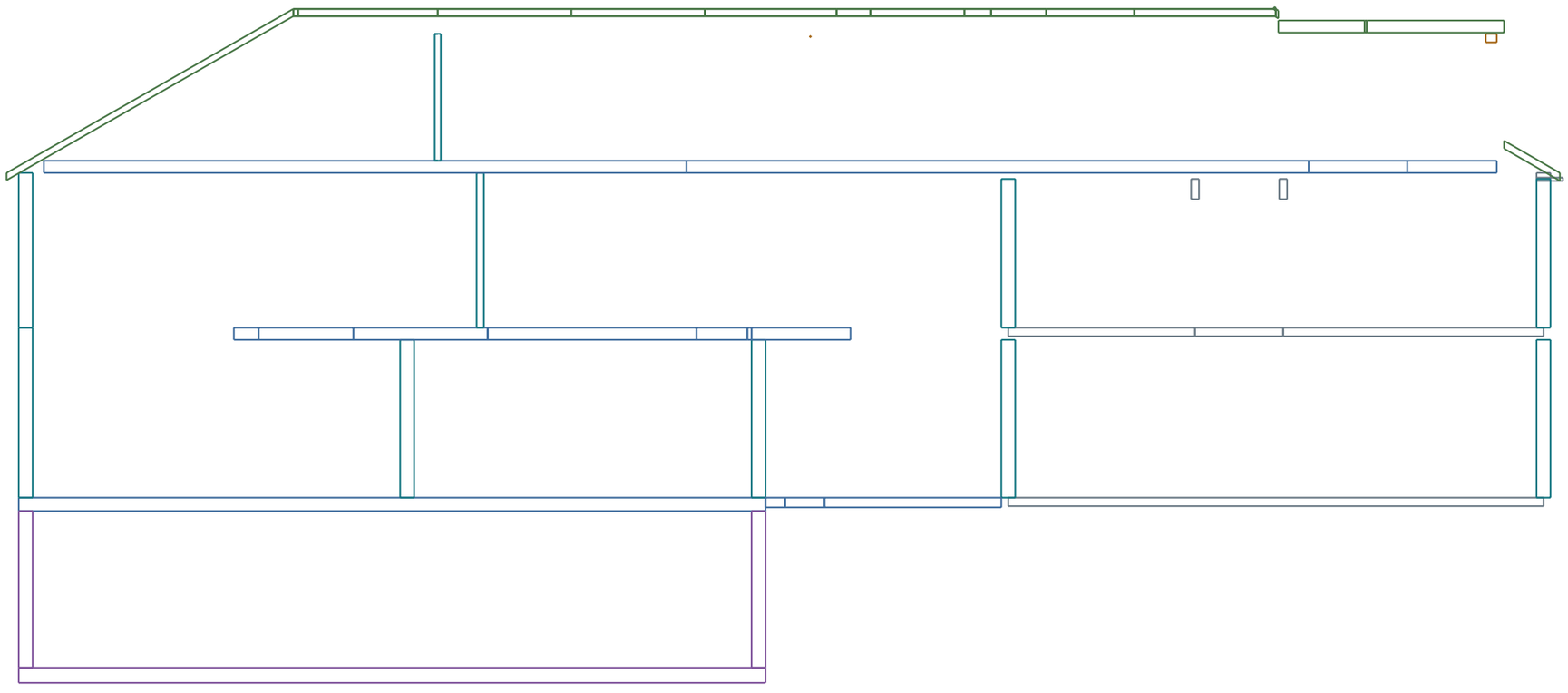
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# Longitudinal coordination section

Shared element geometry / dimensions in model metres / use the interactive register to identify lines



5 m · model geometry

## Reading this drawing

Cyan: wall envelopes.

Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

Exact mesh cut at model X = 9.455 m. No below-ground foundation geometry beyond the architectural basement base is inferred.

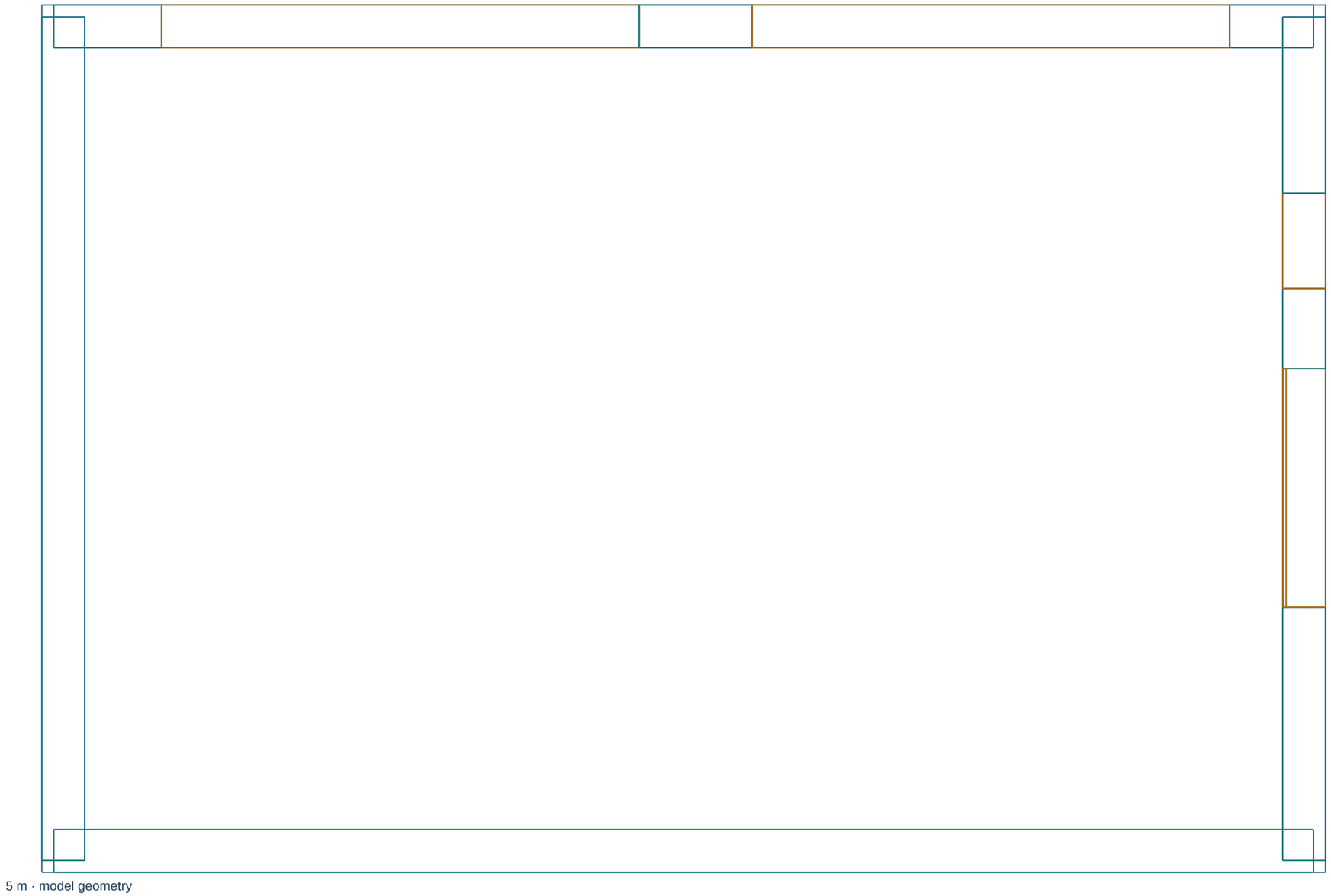
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# Garden workshop coordination plan

Shared element geometry / dimensions in model metres / use the interactive register to identify lines




## Reading this drawing

Cyan: wall envelopes.

Amber: modelled opening-head candidates.

Blue: floor envelopes, not a joist layout.

Green: roof envelope zones.

Purple: basement shell / foundation boundary.

Coral dashed: named support studies.

Grey: retained architectural elements.

Plans cut at 1.20 m above the selected floor. Floor outlines and opening heads are projected for coordination. Hidden/interior lines remain visible.

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# Evidence and reference register

Source geometry, issue identity and independent guidance

## Model source

outputs/output-proposed-compact/geometry.json
outputs/output-proposed-compact/navigation.json
These are saved architectural exports. They are not a structural measured survey.

## Source fingerprint

Geometry SHA-256:
3fb81b911061a35f7b4903f8f7d867fd
5aeafc029de295dd7e00dc152f7b87b7

## Live element register

The local workspace is available at http://127.0.0.1:8765/structure/. Its model.json and schedule.json retain complete element references and source hashes. Geometry changes invalidate the structural build rather than silently leaving a stale register.

## England - structural guidance

Approved Document A: structure
Foundations, walls, floors and roofs; checked 26 October 2026.

## Ground and moisture

Approved Document C
Ground conditions and moisture guidance; checked 26 October 2026.

## Domestic structural engineering

IStructE: introduction for domestic clients
Structural assessment, calculations, foundation investigation and on-site measurement; checked 26 October 2026.

## Scope of this issue

Main house and garden workshop architectural bodies plus named support studies. Landscape, pool, pavilion, decorative layers and services excluded. Modelled head/lintel candidates only; not every structural opening has been reconciled.